3.2 Law of Cosines

Pre-Calculus 3.2 · Student sheet

Name _____ Period ____ Date _____

I can…

- use the Law of Cosines to solve oblique triangles (SSS or SAS).
- decide whether to use the Law of Sines or the Law of Cosines.
- use Heron's Area Formula and solve real-life problems.

Vocabulary · fill in the missing word

Law of Cosines — $a^2 = b^2 + c^2 - 2bc \cos A$ (and the same _____ for b^2 and c^2)

Heron's Area Formula — Area = _____, where $s = \frac{a + b + c}{2}$

Semiperimeter — s = _____

Part 1 · What if it isn't a right angle?

On blank paper, draw three triangles with sides a = 7 cm and b = 8 cm and the angle C between them (use a ruler and a protractor). Measure the third side c to the nearest tenth of a centimeter.

Angle C	c (measured)	c^2	$a^2 + b^2$	Difference
60°			113	
90°			113	
120°			113	

Use a calculator: what is 2ab cos C for C = 60° and for C = 120°? _____

Part 2 · Notes: fill in the blanks

Law of Cosines: c^2 = _____

To find an angle: cos C = _____

Use the Law of Cosines first for _____ and _____.

For SSS, find the _____ angle first.

Heron's Area Formula: s = _____, Area = _____

Part 3 · Examples (work them with the class)

Example 1

Solve the triangle: a = 6, b = 8, c = 12. Round to two decimal places.

Example 2

Solve the triangle: A = 115°, b = 15, c = 10. Round to two decimal places.

Example 3

Which law would you use first?

a. $A = 40°, B = 70°, c = 12$ b. $a = 9, b = 11, C = 52°$ c. $a = 4, b = 6, c = 7$ d. $A = 35°, a = 10, b = 14$

Example 4

Find the area of a triangle with sides $a = 43$ m, $b = 53$ m, and $c = 72$ m.

Example 5

On a baseball diamond, the bases are 90 feet apart and the pitcher's mound is 60.5 feet from home plate on the line to second base.

How far is the pitcher's mound from first base?

Part 4 · Partner practice

Partner 1

A triangle has sides 5, 7, and 10. Find its largest angle. Round to two decimal places.

Partner 2

$C = 40°, a = 9, b = 12$. Find c. Round to two decimal places.